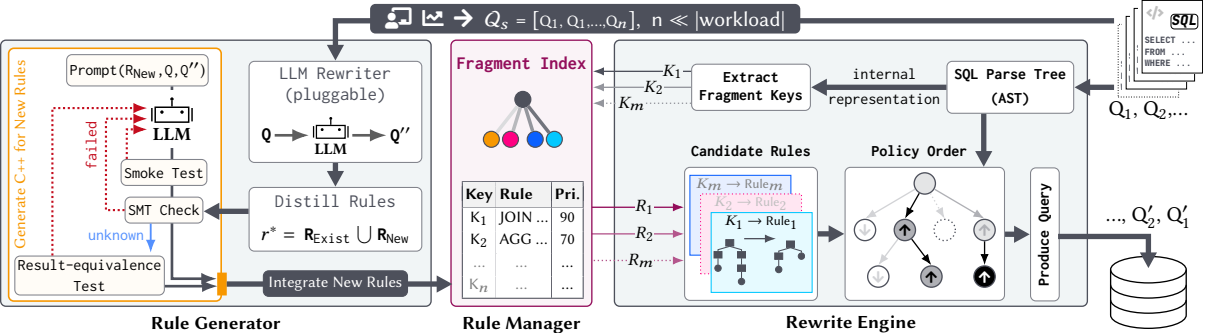

Q_s = [Q_1, Q_1,...,Q_n], n ≪ |workload|
SQL
SELECT ... FROM ... WHERE ...
Q_1, Q_2,...
Generate C++ for New Rules
Prompt(R_{New},Q,Q'')
LLM
failed
Smoke Test
SMT Check
unknown
Result-equivalence Test
LLM Rewriter (pluggable)
Q → LLM → Q''
Distill Rules
$r^* = R_{Exist} \bigcup R_{New}$
Integrate New Rules
Rule Generator
Fragment Index
Key | Rule | Pri.
K_1 | JOIN ... | 90
K_2 | AGG ... | 70
K_n | ... | ...
Rule Manager
K_1
K_2
K_m
Extract Fragment Keys
internal representation
SQL Parse Tree (AST)
Candidate Rules
$K_m \rightarrow Rule_m$
$K_2 \rightarrow Rule_2$
$K_1 \rightarrow Rule_1$
Policy Order
R_1
R_2
R_m
Produce Query
..., Q'_2, Q'_1
Rewrite Engine